Kruskal's Algorithm (Analysis).

Ackermann Function:

$$A(m,n) = \begin{cases} n+1 & m=0 \\ A(m-1,1) & m>0,\ n=0 \\ A(m-1, A(m,n-1)) & m>0,\ n>0 \end{cases}$$

"fast growing function".

$A(0,n) = n+1.$

$A(1,n) = n+2.$

$A(2,n) = 2n+3.$

$A(3,n) = 2^{n+3} - 3.$

Define $\alpha(n)$ = the smallest hierarchy level sufficient to reach n.

↑ Inverse

(slow growing function).

Variation of Ackermann Function (VAF):

For integers $k>0$, $j\geq 1$ (k- level function of VAF)

$$A_k(j) = \begin{cases} j+1 & \text{if } k=0 \\ A_{k-1}^{(j+1)}(j) & \text{if } k\geq 1 \end{cases} \quad \text{— (1)}$$

$$A_{k-1}^{(0)}(j) = j \quad \text{— (2)}$$

$$A_{k-1}^{(i)}(j) = A_{k-1}\left(A_{k-1}^{(i-1)}(j)\right) \quad \text{— (3)}$$

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Lemma 4.

(i) $A_0^{(i)}(j) = j+i$

(ii) $A_1(j) = 2j+1$

Proof (by induction on i).

(i) Base case : $i=0$

Put $k=1$ in ②

$$A_0^0(j) = j = j+0$$

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Hence result holds for $i=0$.

Induction hypothesis: Assume the result holds for $i-1$.

That is, $A_0^{(i-1)}(j) = j+(i-1)$

$$A_0^{(i)}(j) = A_0\left(A_0^{(i-1)}(j)\right) \qquad \text{(Put } k=1 \text{ in (3))}$$

$$= A_0\left(j+(i-1)\right) \qquad \text{(By Induction hypothesis)}$$

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$$= \left(j + (i-1)\right) + 1 \qquad \text{(using ①)}$$

$$= j + i$$

Hence the result holds by induction.

(ii) To prove: $A_1(j) = 2j+1$

$$A_1(j) = A_0^{(j+1)}(j) \qquad \text{(using ①)}$$

$$= j + (j+1) \qquad \text{(By Lemma 4(i))}$$

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$$= 2j+1 .$$

Lemma 5:

(i) $A_1^{(i)}(j) = 2^i(j+1) - 1$

(ii) $A_2(j) = 2^{j+1}(j+1) - 1$

Proof by induction:

(i) Base-Case $i=0$

(Put $k=1$ in ②)

$$A_1^{(0)}(j) = j^0$$

$$= 2^0(j+1) - 1.$$

Hence result holds for $i=0$.

Induction hypothesis: Assume the result holds for $i-1$ that is $A_1^{(i-1)}(j) = 2^{i-1}(j+1) - 1$.

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$$A_1^{(i)}(j) = A_1\left(A_1^{i-1}(j)\right)$$ (Put $k=2$ in ③)

$$= A_1\left(2^{i-1}(j+1)-1\right)$$ (By induction hypothesis)

$$= 2\left(2^{i-1}(j+1)-1\right)+1$$ (By Lemma 4 (ii))

$$= 2^i(j+1) - 2 + 1$$

$$= 2^i(j+1) - 1$$

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The result holds by induction.

(ii) To prove: $A_2(j) = 2^{j+1}(j+1) - 1$

$$A_2(j) = A_1^{(j+1)}(j) \qquad \text{(Using (1) with } k=2\text{)}$$

$$= 2^{j+1}(j+1) - 1 \qquad \text{(By Lemma 5 (i))}$$

Find $A_0(1)$, $A_1(1)$, $A_2(1)$, $A_4(1)$.

$A_0(1) = 2$

$A_1(1) = 3$

$A_2(1) = 7$

$A_3(1) = 2047$

$A_4(1) = ?$

$A_0(1) = 1+1 = 2$ (Using ① $k=0$)

$A_1(1) = 2(1)+1$ (By Lemma 4(ii))

$= 3$

$$A_3(1) = A_2^{(1+1)}(1) \quad (\text{Using } ①)$$

$$= A_2^2(1)$$

$$= A_2\left(A_2^{(1)}(1)\right) \quad (\text{Using } ③)$$

$$= A_2\left(A_2\left(A_2^{(0)}(1)\right)\right) \quad (\text{Using } ③)$$

$$= A_2\left(A_2(1)\right) \quad (\text{Using } ②)$$

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$$A_2(1) = 2^{1+1}(1+1) - 1 \qquad \text{(By Lemma 5(ii))}$$

$$= 2^2(2) - 1$$

$$= 8 - 1$$

$$= 7$$

$$=$$

$$= A_2(7) \qquad (A_2(1)=7 \text{ found})$$

$$= 2^{7+1}(7+1)-1 \qquad (\text{By Lemma 5 (ii)})$$

$$= 2^8(8)-1$$

$$= 2^{11}-1$$

$$= 2047$$

$$A_4(1) = A_3^{(2)}(1) \qquad (\text{Using } ①)$$

$$= A_3\left(A_3^{(1)}(1)\right) \qquad (\text{Using } ③)$$

$$= A_3\left(A_3\left(A_3^{(0)}(1)\right)\right) \qquad (\text{Using } ③)$$

$$= A_3\left(A_3(1)\right) \qquad (\text{Using } ②)$$

$$> A_2\left(A_3(1)\right)$$

$$= A_2(2047) \qquad (\text{Using } ①)$$

$$= = 2^{2048}(2048) - 1$$

$$= > 2^{2048} = (2^4)^{512}$$

$$= (16)^{512}$$

$$> 10^{512} > 10^{80}$$

(No of atoms in observable universe).

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Inverse of $A_k(n)$

$$\alpha(n) = \min\{k \mid A_k(1) \geq n\}$$

$$\alpha(n) = \begin{cases} 0 & 0 < n \leq 2 \\ 1 & n = 3 \\ 2 & 4 \leq n \leq 7 \\ 3 & 8 \leq n \leq 2047 \\ 4 & 2048 \leq n \leq A_4(1) \end{cases}$$

For practical purposes,

assume $\alpha(n) \leq 4$.

Kruskal's Algorithm

Sorting + Implementation of Disjoint set unions (DSU)

Sorting $\rightarrow \Theta(E \log E) = \Theta(E \log V)$

DSU $\rightarrow \Theta(E \cdot \alpha(V)) \underset{\leq 4}{\approx} \Theta(E)$

Let G be connected graph

$E \rightarrow$ no of edges , $V \rightarrow$ no of vertices

$$E = {}^{V}C_{2}$$

$$= \frac{V(V-1)}{2}$$

$$E \in \Theta(V^2)$$

$$\log E = \log V^2$$

$$= 2 \log V$$

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$\in \Theta(\log V)$

Sorting : $\Theta(E \log V)$

DSU : $\Theta(E)$

Sorting dominates DSU.

Sorting time

Ackmann's Function :

$$A(2,n) = 2n+3$$

$$A(3,n) = 2^{n+3} - 3$$

$$A(4,n) = 2^{2^{\cdot^{\cdot^{\cdot^{2}}}}} - 3 \quad (n+3 \text{ twos})$$

$$A(4,0) = 2^{2^2} - 3 = 65533$$

$$A(4,2) = 2^{2^{65536}} - 3 \approx 10^{10^{20000}}$$

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